

Lecture 9 : Differentiation

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1. Sign of the derivative

Let $I \subseteq \mathbb{R}$ be an interval and a function $f : I \rightarrow \mathbb{R}$.

Let c be an interior point of I . Then,

1. function f is said to be **increasing function** at point c , if there exists a positive number δ , such that $f(x) < f(c)$ for all $x \in I$, satisfying $c - \delta < x < c$ and $f(x) > f(c)$ for all $x \in I$ satisfying $c < x < c + \delta$,
2. function f is said to be **decreasing function** at point c , if there exists a positive number δ , such that $f(x) > f(c)$ for all $x \in I$, satisfying $c - \delta < x < c$ and $f(x) < f(c)$ for all $x \in I$, satisfying $c < x < c + \delta$.

Let c be the left end point of I .

1. function f is said to be **increasing function** at point c , if there exists a positive number δ , such that $f(x) > f(c)$ for all $x \in I$, satisfying $c < x < c + \delta$,
2. function f is said to be **decreasing function** at point c , if there exists a positive number δ , such that $f(x) < f(c)$ for all $x \in I$, satisfying $c < x < c + \delta$.

Theorem 1 (*Darboux Theorem*)

Let $I = [a, b]$ and a function $f : I \rightarrow \mathbb{R}$ be differentiable on I . Let $f'(a) \neq f'(b)$. If k be a real number lying between $f'(a)$ and $f'(b)$, then there exists a point c in (a, b) , such that $f'(c) = k$.

Proof: Without loss of generality, let $f'(a) < k < f'(b)$.

Let us define $g : [a, b] \rightarrow \mathbb{R}$ by $g(x) = f(x) - kx$, $x \in [a, b]$.

g is differentiable on $[a, b]$ and therefore g is continuous on $[a, b]$. Consequently, g will attain the minimum value (the greatest lower bound) at some point c in $[a, b]$.

$g'(a) = f'(a) - k < 0$. This implies, function g is decreasing at a . Therefore, there exists a positive number δ , such that $g(x) < g(a)$ for all $x \in [a, b]$, satisfying $a < x < a + \delta$.

This shows that, $g(a)$ is not the minimum value of g on $[a, b]$.

$g'(b) = f'(b) - k > 0$. This implies, g is increasing at b . Therefore, there exists a

positive number δ , such that $g(x) < g(b)$ for all $x \in [a, b]$, satisfying $b - \delta < x < b$.

This shows that, $g(b)$ is not the minimum value of g on $[a, b]$.

Thus, $c \neq a$, $c \neq b$ and therefore $a < c < b$. Since, $c \in (a, b)$, $g'(c)$ exists. We prove that $g'(c) = 0$.

Let $g'(c) > 0$. Then there exists a positive number δ , such that $g(x) < g(c)$ for all $x \in [a, b]$, satisfying $c - \delta < x < c$. This contradicts that $g(c)$ is the minimum value of g on $[a, b]$. Therefore, $g'(c) \not> 0$.

Let $g'(c) < 0$. Then there exists a positive number δ , such that $g(x) < g(c)$ for all $x \in [a, b]$, satisfying $c < x < c + \delta$. This contradicts that $g(c)$ is the minimum value of g on $[a, b]$. Therefore $g'(c) \not< 0$.

Consequently, $g'(c) = 0$, i.e., $f'(c) = k$ and the theorem is established.

Example : Let $f : [-1, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0 & \text{when } x \in [-1, 0] \\ 1 & \text{when } x \in (0, 1]. \end{cases}$$

Does there exist a function g , such that $g'(x) = f(x)$, $x \in [-1, 1]$?

Solution : If possible, let there exist a function $g : [-1, 1] \rightarrow \mathbb{R}$, such that $g'(x) = f(x)$, $x \in [-1, 1]$.

Then, g is differentiable on $[-1, 1]$ and

$$g'(x) = \begin{cases} 0 & \text{when } x \in [-1, 0] \\ 1 & \text{when } x \in (0, 1] \end{cases}$$

Since, g is differentiable on $[-1, 1]$ and $g'(-1) \neq g'(1)$, by Darboux's theorem g' must assume every real number lying between $g'(-1)$ and $g'(1)$, i.e., between 0 and 1 on $[-1, 1]$. But this is not so and therefore g does not exist.

Theorem 2 (Rolle's theorem)

Let a function $f : [a, b] \rightarrow \mathbb{R}$ such that

1. f is continuous on $[a, b]$,
2. f is differentiable at every point of (a, b) , and
3. $f(a) = f(b)$.

Then there exists at least one point ξ in (a, b) , such that $f'(\xi) = 0$.

Proof: Since, f is continuous on $[a, b]$, f is bounded on $[a, b]$.

Let $\sup_{x \in [a, b]} f(x) = M$, $\inf_{x \in [a, b]} f(x) = m$.

By the property of continuity there exists a point c in $[a, b]$, such that $f(c) = M$ and there exists point d in $[a, b]$, such that $f(d) = m$.

Two cases arise.

1. $M = m$.

In this case $f(x) = M$ for all $x \in [a, b]$. Therefore, $f'(x) = 0$ for all $x \in [a, b]$. The theorem holds trivially in this case.

2. $M \neq m$.

In this case at least one of M and m , if not both, must be unequal to $f(a)$ (and $f(b)$).

Let $M \neq f(a)$. Then, $c \neq a$, $c \neq b$ and therefore $a < c < b$.

By condition (2), $f'(c)$ exists.

If possible, let $f'(c) > 0$. Then, $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} > 0$.

Hence, there exists a positive number δ , such that $\frac{f(x) - f(c)}{x - c} > 0$ for all $x \in [a, b]$, satisfying $0 < |x - c| < \delta$.

This implies, $f(x) - f(c) > 0$ for all $x \in [a, b]$ satisfying $c < x < c + \delta$.

i.e., $f(x) > M$ for all $x \in [a, b]$, satisfying $c < x < c + \delta$.

This contradicts that, M is the supremum of f on $[a, b]$.

Consequently,

$$f'(c) \not> 0. \quad (1)$$

If possible, let $f'(c) < 0$. Then, $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} < 0$.

Hence, there exists a positive number δ , such that $\frac{f(x) - f(c)}{x - c} < 0$ for all $x \in [a, b]$, satisfying $0 < |x - c| < \delta$.

This implies, $f(x) - f(c) < 0$ for all $x \in [a, b]$ satisfying $c - \delta < x < c$.

i.e., $f(x) < M$ for all $x \in [a, b]$, satisfying $c - \delta < x < c$.

This contradicts that, M is the supremum of f on $[a, b]$.

Consequently,

$$f'(c) \not< 0 \quad (2)$$

From (1) and (2) it follows that $f'(c) = 0$. Therefore, $\xi = c$.

If however, $M = f(a) = f(b)$, then $m \neq f(a)$ and therefore $d \neq a$, $d \neq b$ and $a < d < b$.

Proceeding with similar arguments, we can prove $f'(d) = 0$. Therefore, $\xi = d$.

This completes the proof of the theorem.

Example : Verify Rolles theorem for the function:

$$f(x) = (x - 1)(x - 2), \quad x \in [1, 2].$$

Solution : Since, $(x - 1)(x - 2)$ is continuous in the interval $[1, 2]$. Also,

$$f'(x) = 2x - 3$$

since $f(x)$ is differentiable in $(1, 2)$ and $f(1) = 0 = f(2)$.

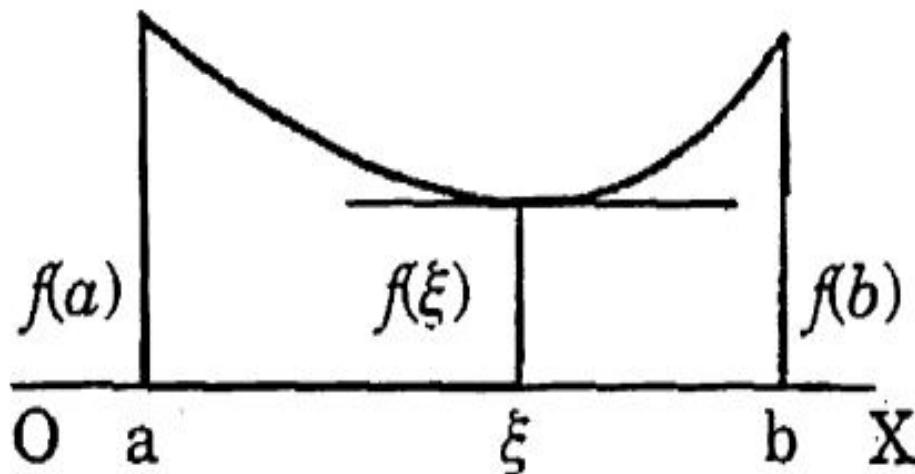
This implies that, Rolles theorem is applicable here. Then, by Rolles theorem there must be at least one c , such that $f'(c) = 0$.

$$\implies f'(c) = 2c - 3 = 0 \implies c = \frac{3}{2}.$$

As, $c = \frac{3}{2} \in [1, 2]$, so at $c = 3/2$, $f'(c) = 0$.

Geometrical Interpretation

If a function f has a graph which is a continuous curve on the interval $[a, b]$ and the curve has a tangent at every point on it with abscissa between a and b and the ordinates $f(a)$, $f(b)$ are equal, then there exists at least one point ξ , in (a, b) such that the tangent to the curve at $(\xi, f(\xi))$ is parallel to the x-axis.



Examples

Reference book : *Introduction to real analysis by S.K. Mapa.*